# Pack the truck

## Optimization · Level 1 · Lesson 1



# In this lesson you will

- Say the rule and the goal of a packing puzzle.
- Check whether a load keeps the rule.
- Find the best load, and think about what stays behind.

# Key ideas

- The rule: no more than 1,000 kg
- The goal: the most points
- Quick way: 60 points; best: 65 points
- Something always stays behind

# Activity: Pack the bag

1. Cards (5 min): Make six cards with a weight and points.
2. Pack (8 min): Find loads that keep the limit of 10.
3. Best (5 min): Which load has the most points?
4. Share (2 min): Can your partner beat it?

**No cards? Tear a page into six pieces. No device needed.**

# Let’s discuss

1. Why didn’t the quick way give the best load?
2. Why doesn’t a load that is too heavy count?
3. Who should decide what stays behind?

# Exit ticket

- The best truck load is ___ and ___, with ___ points.